

- 1 (a) Give full details of the geometrical transformation in the x - y plane represented by the matrix $\begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$. [1]

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Let $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 2 & 2 \end{pmatrix}$.

- (b) The triangle DEF in the x - y plane is transformed by $\mathbf{A}$ onto triangle PQR .

Given that the area of triangle DEF is 13 cm^2 , find the area of triangle PQR . [2]

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- (c) Find the matrix $\mathbf{B}$ such that $\mathbf{AB} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$. [2]

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- (d) Show that the origin is the only invariant point of the transformation in the x - y plane represented by $\mathbf{A}$. [4]

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